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User perspectives on LINK: A generative AI chatbot for academic libraries

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ABSTRACT

This article presents a case study on the development, phased testing, and user evaluation of Library Information & Knowledge (LINK), a generative AI–enabled library chatbot created by the University of South Florida Libraries. Built using Microsoft Copilot Studio, LINK integrates conversational AI with structured knowledge documents to extend library services, improve efficiencies, and support user navigation of library resources. Development required extensive groundwork, including web content mapping, knowledge base restructuring, and iterative refinement informed by internal staff, faculty, and student testing. An IRB-approved survey captured user perceptions of ease of use, accuracy, usefulness, and trust. Respondents valued LINK's 24/7 availability and conversational flexibility, but noted limitations tied to its ethical design constraints, particularly its inability to access licensed databases. These findings reflect broader trends in the literature, which highlight the tension between user expectations and the actual capabilities of AI chatbots in academic settings. The study identifies priorities for future development, including reducing the database access gap, expanding knowledge coverage, improving accuracy with structured and source-linked responses, and strengthening pathways to human assistance. Despite technical and structural challenges, the project demonstrates how low-code generative AI tools can enhance academic library service ecosystems while reinforcing transparency and user trust.

Introduction

Library chatbots have been investigated and developed by libraries as a way to extend services beyond closing times, alleviate service workloads, and improve patron interactions. The availability of generative artificial intelligence (AI) or large language model (LLM) development platforms in higher education has created new opportunities to build or enhance chatbot models for even greater service enhancement and efficiencies. Generative AI also brings user concerns over ethics, biases, and transparency. The incorporation of a library chatbot where there wasn't one before, or the integration of generative AI into an existing library chatbot, can impact user trust in the library and library reference services. This article explores and analyzes the development of a generative AI enabled library chatbot within a multi-campus academic library setting, with special attention to enhanced transparency and end user feedback. It incorporates the results of an IRB approved study that further explored the experiences and viewpoints of chatbot users and early testers.

Literature review

This case study is rooted in an examination of the state of academic library adoption of chatbots from the early 2010s to 2025. Early adaptations of chatbots in libraries were based on pathed conversations developed by library staff to answer common questions. The bots evolved to incorporate increasingly sophisticated natural language programming where teams could adapt bots to help patrons using conversational phrasing. Early efforts at developing library chatbots sparked in 2012, with systems developed using backends like SQL databases (Allison, 2012) or other structures described by some as "weak AI" (Ratledge, 2012). Literature from this period emphasized funding issues (Kane, 2019; McNeal & Newyear, 2013; Ratledge, 2012) or continuity from staff changes (Allison, 2012) as motivators for adopting chatbots as a 24/7 service option. The University of California, Irvine reported accuracy rates of 70% three years after implementing their system in 2014 (Kane, 2019). With evidence of success, and increased need showcased by events like the COVID-19 pandemic shutdowns, scholars looked to chatbots with AI-enhanced technology to provide a model for how libraries may evolve their reference services (Panda & Chakravarty, 2022).

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As chatbots became more common in libraries, several institutions developed or adopted more advanced open-source or proprietary bots that guided their users through questions to provide resources in an interactive window (Bagchi, 2020; Ehrenpreis & DeLooper, 2022; Reinsfelder & O'Hara-Krebs, 2023; Rodriguez & Mune, 2022; Xuan, 2025). Concerns over risks of inaccuracy, dissolving trust, and ethical issues linked to the use of AI as a reference assistant are persistent threads in the literature (Ehrenpreis & DeLooper, 2022; Guy et al., 2023; Kaushal & Yadav, 2022). Alternatively, Mckie and Narayan (2019) found that the adoption of a chatbot at the University of Technology Sydney was able to reduce library anxiety by helping students to move past the first step of inquiry and prepare themselves for research. They argue, "conversational interfaces must rely on the constitution of artificial personality to establish a consistent user experience," which they accomplished by "designing an empathetic personality and social roles into our chatbots [to] create new characters to guide undergraduate students in their individual journeys through university" (Mckie & Narayan, 2019, p. 274). The level of effort put into customizing personalities within library chatbots could use more examination across the field, but developing or retaining a trust relationship remains a vital aspect of adopting an AI chatbot.

While academic libraries have learned a considerable amount about the potential for chatbots to expedite responses to patron queries and relieve financial and staffing limitations for reference services, few libraries have adopted a model that taps into the potential of generative AI. Within early months of the launch of OpenAI's ChatGPT, several articles and think pieces asserted the potential for integrating generative LLMs into library chatbot technology to improve a range of library services from reference to cataloging to document delivery (Adetayo, 2023a; Idhris & Peter, 2024; Pival, 2023; Safadel et al., 2023). These studies have been complimented by a handful of extensive integrated literature and systematic reviews (Aboelmaged et al., 2024; Agarwal et al., 2022; Balnaves et al., 2024; Harisanty et al., 2024; Yan et al., 2023).

As of mid-2023, Guy, Pival, Lewis, and Groome found that "most of the chatbots used in the library market are essentially interactive FAQs, where keywords entered by the user return suggestions to pages that may answer the query" (Guy et al., 2023, p. 1). This assessment appears to be consistent across many academic libraries in the first half of the 2020s. Citing Harisanty and others, Garg et al. (2024) argue that if libraries can overcome issues linked to awareness, finances, infrastructure, resistance to change, and lack of expertise, then AI could be adopted more broadly and result in an "upswing [for] the library operations in providing efficient and effective services to the users." (Garg et al., 2024, p. 11) Their case study looks at a more niche adoption of a reference chatbot called AgriRef, which provides resources for farmers. By exploring these smaller roll-outs, it may be possible to determine effective workflows. The University of Delaware implemented a distributed model that used the expertise of people across the library, as well as student assistants, to train their chatbot with relevant information (Twomey et al., 2024). Additional training within libraries for library staff can also be supported using the AI models, which Arizona State University (ASU) adopted in 2024 (Yates et al., 2024). The ASU program received dedicated funding to work collaboratively with OpenAI's ChatGPT as they developed "AI personas", which used custom GPTs to create realistic characters to support training for the bot and library staff. Their work illustrates an interconnected nature in how further development of chatbot services can be used to improve the knowledge base and service quality of library staff, rather than acting as a replacement.

Reasons for limited adoption within academic libraries vary, but with a majority of student's using generative AI in their research already (Deschenes & McMahon, 2024), integrating the technology into library chatbots offers a pathway to leverage the technology with peer-reviewed resources. Lombard (2024) illuminates one of the major research hurdles that generative AI chatbots will face is related to the proprietary nature of library databases. Lappalainen and Narayanan (2023) describe how ChatGPT, or similar, generative AI chatbots can also be "tricked" through "prompt injection," which may "result in unreliable or questionable responses and the bot could start performing tasks beyond its intended scope." However, they remain largely optimistic about the potential of the chatbot and its ability to support reference services. Several similar works are also examining how the adoption of generative AI technologies using systems like ChatGPT, Bing, and IBM Watson are low cost, require little technical expertise to implement, and support students by giving them a comfortable space to engage with their research questions (Adetayo, 2023b; Bonal, 2024; Safadel et al., 2023; Shevchuk & Saltykova, 2024).

Background

University of South Florida (USF) is a high-impact, R1 research institution serving over 40,000 students across three campuses in the Tampa Bay area. Known for its commitment to innovation and cross-disciplinary collaboration, USF supports a wide range of research and instructional activities through its network of centers and academic support units. Among these, the USF Institute for Artificial Intelligence + X has played a leading role in advancing AI exploration across fields long before the widespread release of generative AI tools. Early institutional investments and a decentralized approach to faculty autonomy created a flexible environment where instructors could explore generative AI on their own terms, encouraging experimentation in both pedagogy and scholarship.

When generative AI tools like ChatGPT gained public attention in late 2022, USF was well-positioned to respond. Units such as the former Center for Teaching & Learning and the USF Innovative Education rapidly convened workshops and guidance for faculty, while plans were set in motion for a new academic college focused on AI and allied disciplines. Faculty across disciplines began testing AI's potential in their classrooms and research workflows, sparking critical conversations about academic integrity, ethical usage, and student learning outcomes. Simultaneously, students engaged with AI both inside and outside of formal coursework, contributing to a campus-wide atmosphere of inquiry, debate, and adaptation.

The USF Libraries quickly became a central node in this ecosystem of exploration. Library faculty, who operate under a non-tenure model and participate in shared governance, were empowered to take initiative and develop services responsively. Their longstanding culture of experimentation encouraged early engagement with generative AI. As faculty and students began requesting guidance on citation practices, AI evaluation, and research applications, the Libraries responded with curated guides, hands-on workshops, and targeted consultations. Internally, librarians also began piloting AI to streamline workflows such as meeting summaries, content drafting, transcription, and metadata generation, pushing the boundaries of library service innovation.

USF Libraries administrative support for exploring ways generative AI could improve library efficiencies led to the proposal for an AI-powered chatbot, not only as a functional tool to improve access to library support, but also as an opportunity to deepen the Libraries' experimentation with generative technologies. The chatbot project embodied a strategic shift: from positioning the Libraries solely as a resource for librarian guidance on AI, to embracing it as a potential enhancement to infrastructure for library operations. It also aligned with broader institutional trends, where support units like the Libraries and Innovative Education began developing tools that could be deployed equitably, transparently, and sustainably across the university.

LINK AI chatbot background

Development of a generative AI chatbot was proposed in response to the announcement of the USFLibraries Internal AI Grant by a cross-unit team assembled by a subject liaison librarian in the Research and

Instruction unit from personnel already supporting AI literacy efforts. Early collaboration with USF's Research IT team revealed an opportunity to leverage Microsoft Copilot Studio, a low-code platform already supported across campus. This allowed for rapid prototyping without the need for extensive custom coding. The cross-unit team undertaking the project was comprised of three faculty librarians with expertise in Digital Humanities, Copyright & Intellectual Property, and Research Applications of AI, alongside two staff members from Library Services and the DATA department. Diverse expertise within the team brought broad perspectives and quickly gained support across the Libraries.

The proposal was selected for funding in October 2024, receiving approximately $12,000 to support the pilot phase. Development began with internal data curation, initial prototyping, and broad stakeholder engagement. Recognizing the need for change management, the team conducted focus groups, live walkthroughs, and iterative usability testing. These sessions helped refine the chatbot's core conversational model and ensured it could interpret user intent while maintaining academic credibility.

Prior to development, the team conducted a literature review on library chatbot usage. While many academic libraries had adopted chatbots for reference support, challenges persisted around trust, interaction quality, and transparency. To address these concerns, the team proposed a generative chatbot capable of producing dynamic, natural-sounding responses that transparently linked to their sources—an essential feature for academic contexts. This emphasis on conversational clarity and citation transparency became a defining principle of the project.

Microsoft Copilot Studio was chosen for its integration with Microsoft 365 and its institutional endorsement as a cost-effective, low-code development tool. However, the decision to adopt a "citizen developer" model, as recommended by Research IT, meant the burden of design and implementation fell largely on librarians whose primary assignments did not usually include programming. While the citizen developer model of development dramatically reduced costs and accelerated initial deployment, it introduced significant constraints. Support was minimal, and documentation for advanced features was often sparse or outdated. The team relied heavily on peer troubleshooting, community forums, and iterative testing to work around these challenges.

Despite these limitations, Copilot Studio provided a valuable on-ramp for the Libraries to experiment with generative AI in a hands-on, sustainable way. Its modular interface enabled the team to design a hybrid chatbot that could handle both structured queries and more open-ended academic questions. Although fine-tuning generative behavior and enforcing consistent citation links required creative workarounds, the project successfully demonstrated how libraries can leverage institutionally supported tools to build innovative, mission-aligned services without waiting for perfect infrastructure or external development support. The chatbot, named Library Information & Knowledge (LINK) and known internally as LINK, now stands as a case study in rapid, pragmatic AI deployment within academic libraries.

Foundational planning, data collection, and web mapping

At the start of development, the Library Information & Knowledge (LINK) team conducted significant groundwork to ensure that the model would be both accurate and sustainable. In late Fall 2024, data collection efforts focused on consolidating services, resources, and contact information from across the library. This required tracking down scattered and inconsistently documented information and creating an organizational structure with uniform categories. The AI model's success depended on well-structured knowledge sources, which meant each entry had to include relevant links, clear descriptions, and internally consistent language. The team found that repetitive prompting helped reinforce link-based responses, improve transparency, and enhance discoverability.

A major roadblock emerged early on: the library website lacked a current, comprehensive sitemap. Without a clear view of the library's web presence, the team struggled to determine what information the AI could even access. This led to a formal web mapping initiative, including an internal audit of all library webpages—both public and hidden. The audit revealed numerous gaps, outdated pages, and navigation issues that made key content difficult to find or connect. In response, the team proposed updates, removals, and redesigns to align the site more closely with the AI's logic needs. While the Libraries had staff overseeing web content, turnaround times for edits varied, and the mapping process proved essential to making LINK viable.

Additionally, this early planning stage included an extensive review and restructuring of knowledge documents. These documents were critical to the AI's performance, and poor-quality inputs were quickly identified as a major source of failure during early alpha testing. The team aggressively revised this content, stripping away duplicative or unclear sections and reformatting materials to optimize search and response behavior within Copilot Studio's architecture.

Development timeline and methodology

The development of LINK followed a carefully phased approach, rooted in strong project planning, iterative testing, and responsive adaptation to challenges. Initial planning and preparation began in November and December 2024, during which the team scoped the project, identified core use cases, and evaluated the capabilities of Microsoft Copilot Studio. This early stage also involved setting goals for ethical deployment, stakeholder buy-in, and long-term scalability.

In January 2025, the team entered the pre-alpha buildout phase. During this stage, they constructed a foundational logic loop that balanced generative AI responses with structured decision paths, knowledge document queries, and fail-safes. Early development revealed critical gaps—many relevant library resources were siloed or inaccessible to the AI model, including LibGuides, database content, and even structured tables on library webpages. To mitigate this, the team decided to limit the model's training to library-authored content only, eliminating reliance on general internet information. This decision helped reduce hallucinations and ensured that responses remained rooted in authoritative, USF-specific sources.

The alpha development phase, spanning February to May 2025, involved intensive refinement of the model's logic and content structure. During this time, internal testing and stakeholder feedback uncovered significant flaws in both the AI's knowledge and the overall organization of library web content. For example, outdated or incomplete webpages often led to failed AI responses, prompting a major overhaul of the knowledge documents. The team worked closely with various departments to standardize service descriptions, contacts, and policies. They also discovered that many advanced Copilot Studio features lacked documentation, requiring creative problem-solving and custom workarounds.

By June 2025, LINK entered beta deployment, running 24/7 while AskALibrarian, the Libraries' long-standing librarian chat service, remained available during regular business hours. This beta phase prioritized user exposure and feedback loops. Weekly updates addressed emergent issues such as faulty logic, missing links, or vague responses. Despite a drop in direct user feedback compared to focus group periods, internal transcript reviews and chatbot problem reports remained essential for tracking down errors and knowledge gaps.

Budget breakdown

The total cost of the LINK project remained modest, with strategic investments directed toward software access, outreach, and publication. Key expenditures included:

- $7123.92 for software infrastructure: This covered two Microsoft 365 licenses, the Copilot Studio deployment itself, and one year of usage credits. It also included setup and launch support from USF Research IT, whose teams had prior Copilot experience.

- $1560 for travel expenses: This funded conference participation for sharing LINK's development as a case study, promoting broader knowledge exchange and visibility.
- $3860 in article processing charges (APCs): Reserved for open access publication in a preferred peer-reviewed journal, ensuring the project's findings remain publicly available and citable.

Testing and community engagement

A core strength of the LINK project was its emphasis on iterative feedback from multiple stakeholder groups. Over the course of development, the team conducted three rounds of structured testing: 1) internal, USF Libraries' staff, 2) external, USF faculty, and 3) external, USF students. Each group played a distinct role in surfacing functionality gaps, usability patterns, and expectations around AI in academic spaces.

The internal library focus group was held in February 2025 as a hands-on event hosted in a USF Libraries instructional lab. Participant personnel from all units in the Libraries were invited to "break the bot" and provide open-ended feedback on both interface and performance. This session was crucial in revealing systemic gaps in the library's own documentation, which led directly to updates in service descriptions and knowledge architecture. Participants also demonstrated novel ways of asking questions, phrasing queries differently than the bot anticipated, which led to deeper improvements in pattern recognition and response paths. Refreshments were provided, and informal conversations yielded some of the most actionable insights.

Fifteen USF faculty members participated in both virtual and in-person sessions in March of 2025. Faculty testing focused less on individual queries and more on the ethics and expectations surrounding AI deployment. Their feedback centered on broader capabilities. Some participants assumed LINK would be able to access user accounts, search the catalog, or provide personalized reading lists. These assumptions revealed a need for better messaging about the tool's scope. Faculty appreciated the commitment to transparency and the emphasis on sourcing, and their feedback helped refine how LINK communicates its own limitations. A full lunch was provided for the in-person group to encourage deep discussion.

Student testing proved more challenging to coordinate. Due to event timing falling at the end of the Spring 2025 semester, an in-person event wasn't feasible, so testing was conducted asynchronously. Interested students were emailed access to the chatbot and a feedback form. Although only a subset returned detailed input, those who did provided valuable data. Their queries, often informal or exploratory, helped the team identify both strengths and blind spots in how LINK addressed undergraduate needs. Testing with students reinforced the importance of surfacing key services (e.g., printing, textbook reserves) more directly within the chatbot logic.

In addition to structured events, the development team continuously tested the bot throughout the alpha and beta periods. A student developer with a computer science background worked 20 hours per week to support testing and build-out of focused topics. Their role was instrumental in identifying logic gaps, restructuring prompts, and stress-testing the model across dozens of use cases. The team also maintained a live ticket system and review log, allowing them to address reported issues from logic errors to hallucinations and missing links. Collectively, this multi-pronged testing strategy was critical, not just for debugging, but for fostering buy-in from stakeholders, ensuring that LINK reflected the priorities and workflows of its intended user base.

IRB study on model use

An IRB approved study ran parallel to the testing and launch of the LINK AI Chatbot. The study would survey participants of the planned focus groups on their interaction with the chatbot. Study participants would also be asked for their views on AI technology separate from the chatbot testing.

Methodology

The survey was sent out in two distributions. The first was timed after the first and second focus groups consisting of USF Libraries personnel and USF faculty from all departments, respectively. It was sent to 25 individual emails on March 31st with a reminder at 2 weeks. The second survey distribution was sent to 27 individual emails on June 11th with a reminder at 2 weeks after the student and others focus group which was organized in an asynchronous manner.

The survey consisted of 17 questions, one of which was a Likert with 11 different statements. The majority of questions were multiple choice. Information such as discipline, technology use, library use, and preconceptions of generative AI were gathered by the survey mechanism as well to track how participants background effected their interactions with the chatbot. Since the emails collected during the focus groups and testing period of the LINK AI Chatbot would also be used to share information about chatbot upgrades, new testing opportunities and more, the participants were given the option to opt out of all LINK AI Chatbot interaction at the end of the survey.

Limitations

The potential respondent pool was limited by the number of volunteers that signed up for any one of the three focus groups. The small sample size could limit the ability of the survey responses to provide any kind of generalizable insight. No question was marked as required, so some questions received less response than others. Despite these limitations, the team identified several potential trends to further explore following review of the survey.

Results

The survey received 11 complete responses. A total of 11 respondents answered the question about their role at the university:

- 4 were staff or administrative personnel,
- 4 were faculty,
- 2 were undergraduate students, and 1 graduate student.

In reporting their college or department, 3 selected 'other,' 2 were from the School of Information, 2 were from the College of Artificial Intelligence, Cybersecurity, & Computing, 2 were from the English department, and one was from the School of Theatre and Dance.

All 11 respondents answered the question about which technologies they used on a monthly basis. They were able to select more than one option. The most common technology used was social media recommendations/algorithms (n = 8). Virtual assistants like Siri and smart home devices were the next most popular technologies with 6 selections each. These were followed by smartwatches or other wearables, recommendation systems, and security cameras (Fig. 1).

To the question 'how often to do you use the library's online resources' the most common answer was less than once a week (n = 4), followed by daily and several times a week each with 3 respondents (fig. 2).

The most used library resource reported by respondents was e-books (n = 10) and digital collections (n = 8) followed closely by research guides, special collections, and borrowing physical materials (fig. 3).

When asked about their primary goal for using the library AI chat assistant, both 'getting help with a specific research query' and 'learning more about library services' were the most common response, each chosen by 4 respondents. Finding quick answers to basic questions, receiving personalized recommendations for resources, and other all received two responses. Those respondents who chose other elaborated that 1) they were helping to test it to make it better and 2) they wanted to be aware of a tool their students would likely use.

Respondents were asked about their initial impression of the library AI chat assistant. Most responses were positive, noting the possibilities if the chat assistant continued to be improved. The most frequent criticism



Fig. 1. Technologies commonly used by respondents.

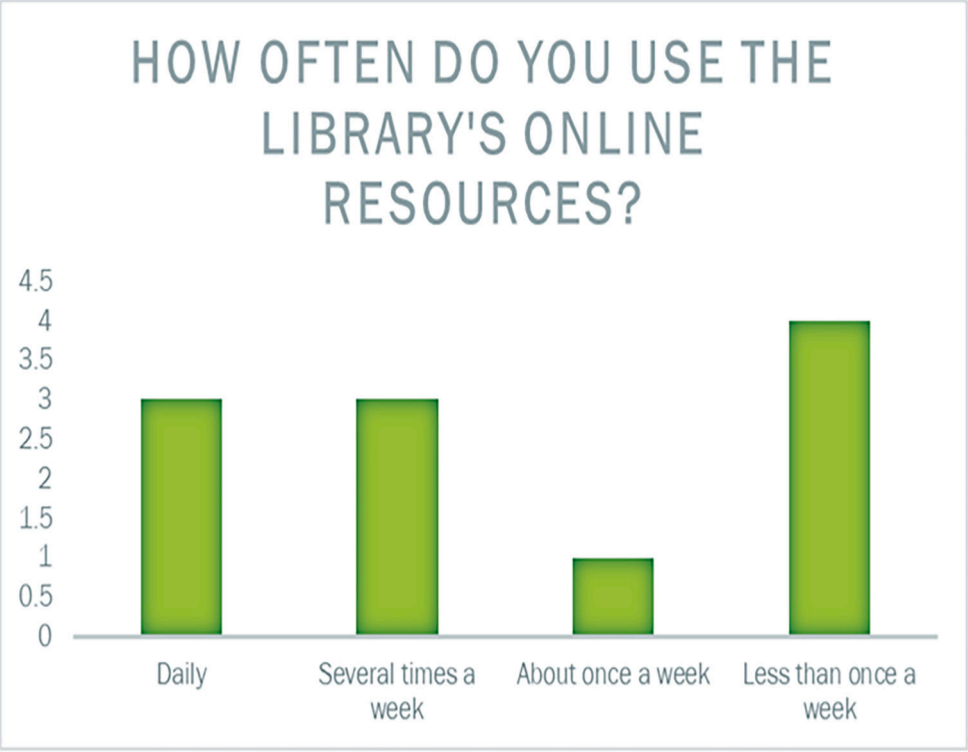


Fig. 2. Online library resource use frequency.

related to the chat assistant's inability to search databases or retrieve answers from the library's holdings.

The respondents were asked to rate their agreement with a selection of 10 statements regarding their experience with the library AI chat assistant (Fig. 4). The most positive responses were associated with the statements 'the library AI chat assistant was easy to use' and 'I routinely use other automated or AI assistants.' Many of the other statements about the accuracy and usefulness of the library AI chat assistant averaged neutral to negative.

The features of the AI chat assistant that respondents found most helpful were 24/7 availability ($n = 9$) and the ability to ask follow-up question ($n = 7$) (fig. 5). Respondents were also asked what they would like to see improved in the AI chat assistant, more comprehensive knowledge and accurate answers were the primary suggestions (Fig. 6).

The remaining questions in the survey solicited open ended responses to a number of questions that would help the development team improve the AI chat assistant for the next version release. Respondents were asked for situations when the chat assistant got things wrong and where was the greatest area for improvement. The primary failings reported for the AI chat assistant were its inability to access databases and provide specific resources on request or for specialized inquiries.

Ability to access databases and other tools was also reported in response to questions about desired added functionality. Most of the responses to the open-ended feedback question were positive and encouraging the further development of the chatbot. The open-ended questions were coded for positive, neutral, and negative tone and

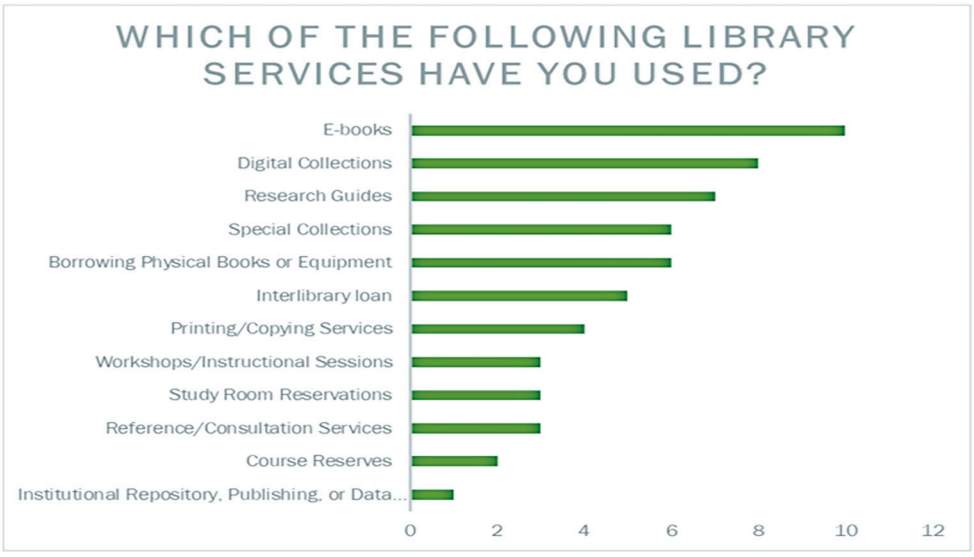


Fig. 3. Library services used by respondents.

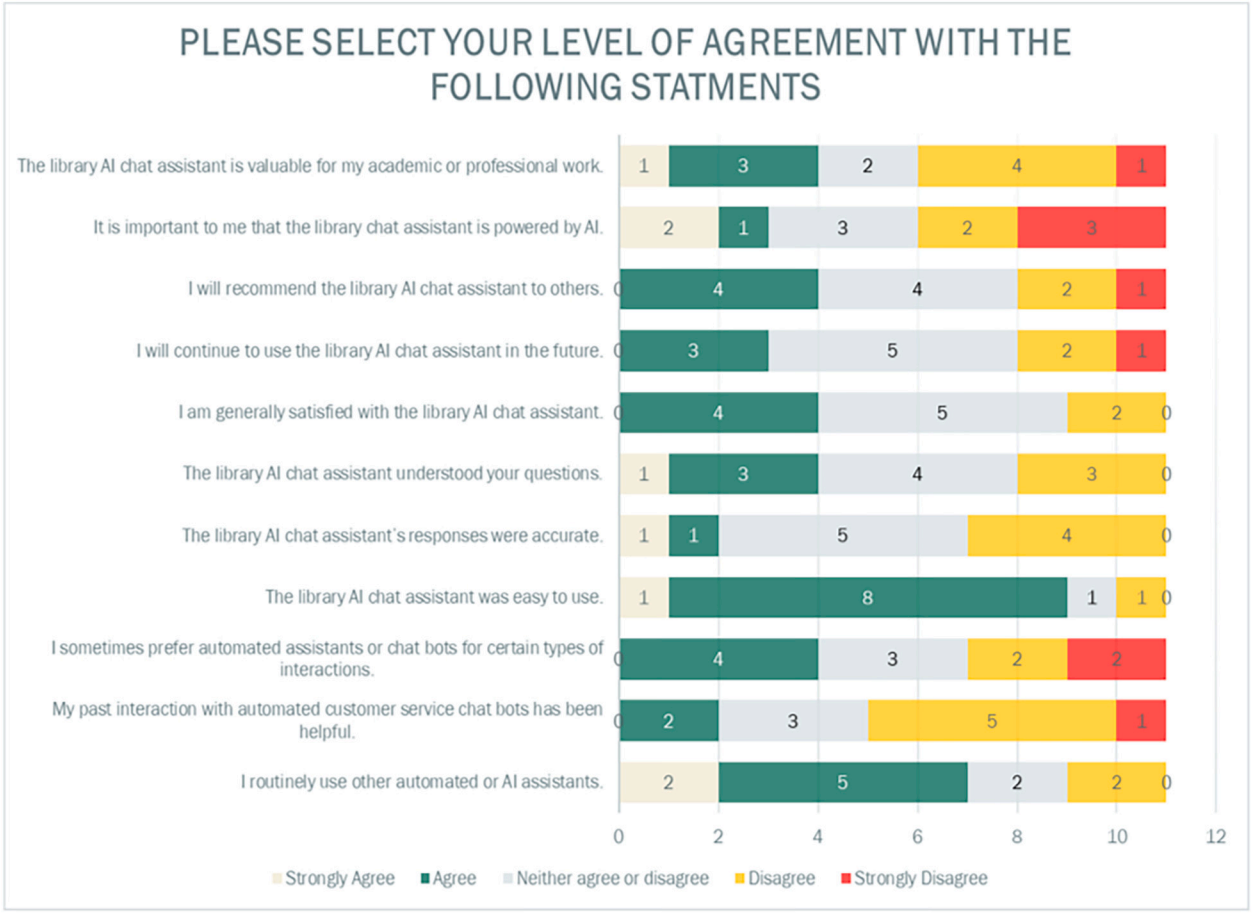


Fig. 4. Respondent answers to Likert scale statements.

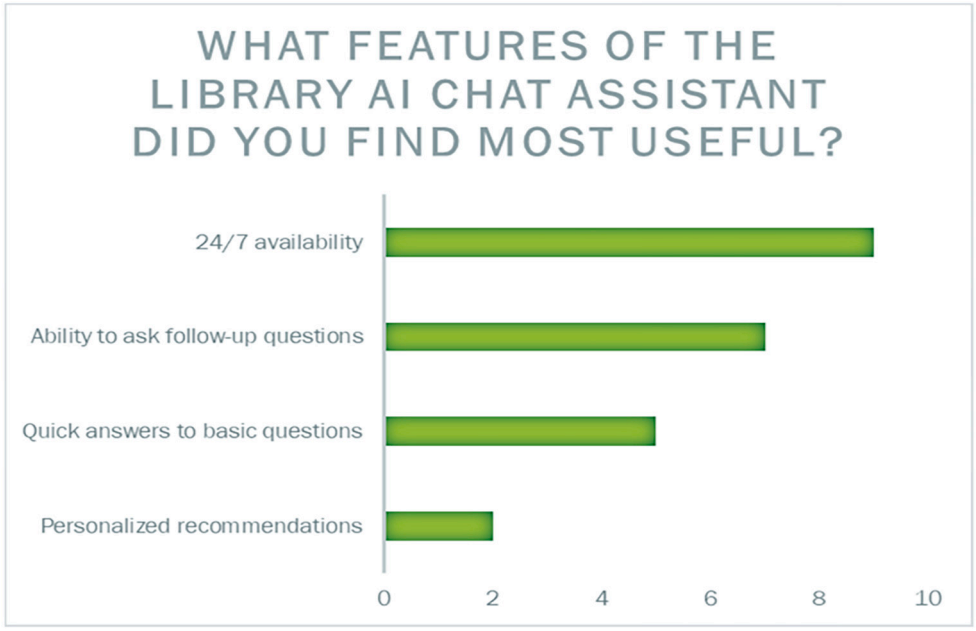


Fig. 5. Chat features found most useful by respondents.

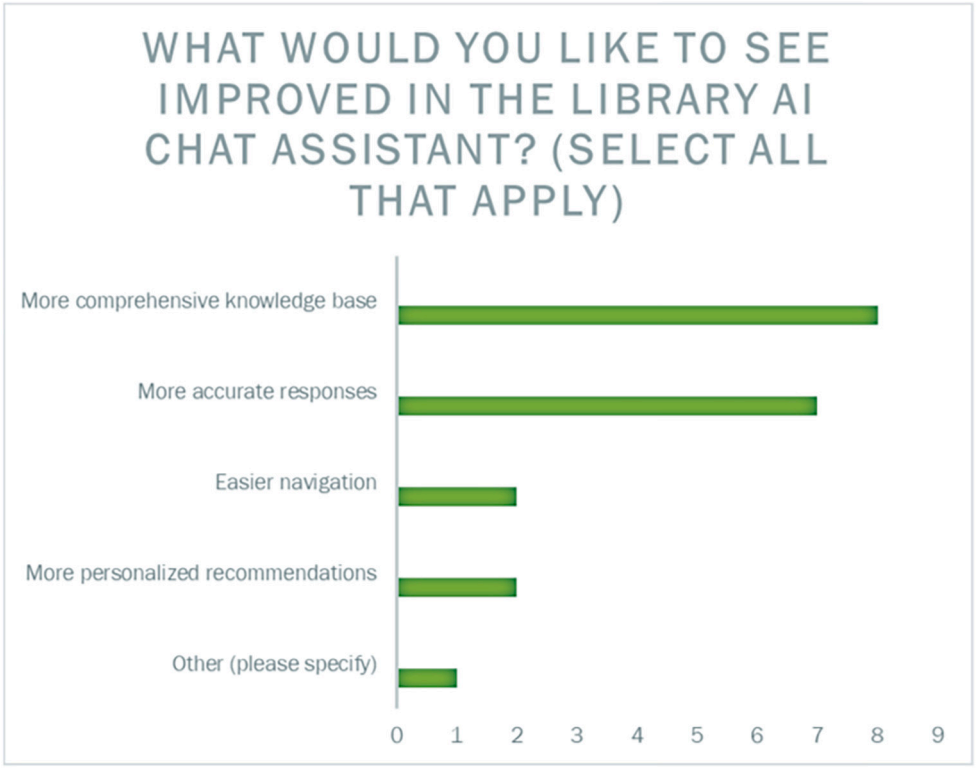


Fig. 6. Suggested improvements.

tabulated based on the nature of the respondent's college or department (Fig. 7).

Discussion

This pilot study of the LINK AI chatbot at USF Libraries offers a window into its current strengths and limitations. Ease of use emerged as the standout positive, with 81.8% of surveyed focus group participants agreeing or strongly agreeing that the chatbot was easy to use. Two features were especially valued: 24/7 availability (81.8%) and the ability to ask follow-up questions (63.6%). These align with prior literature showing that library AI tools are appreciated most for constant availability and conversational continuity, particularly for routine or directional inquiries.

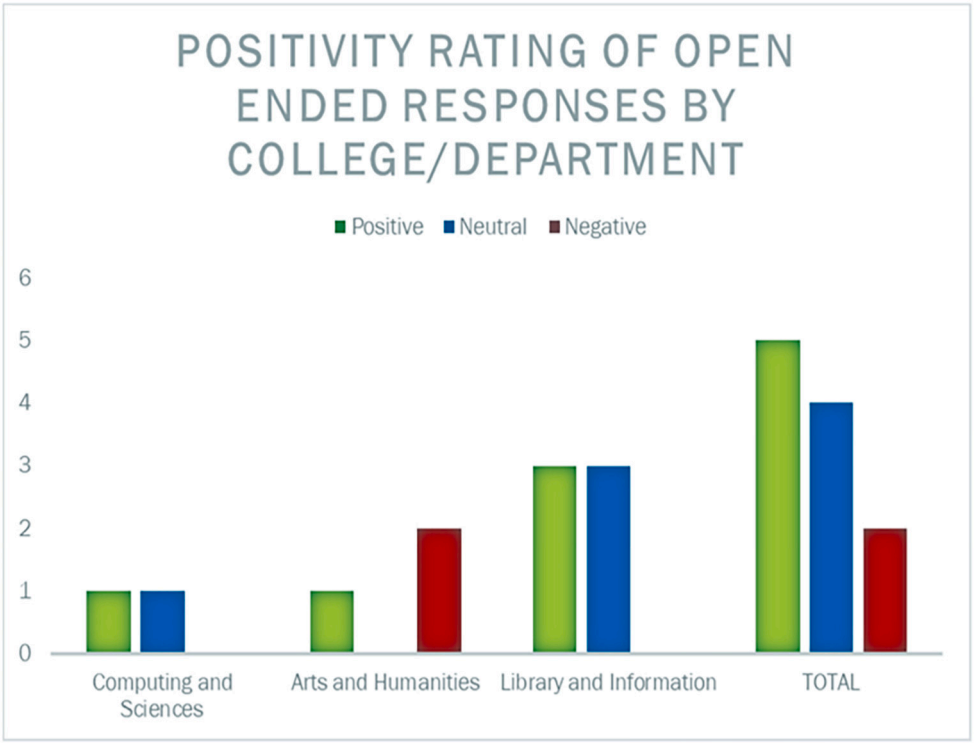


Fig. 7. Positivity rating of open-ended responses by college/department.

The participant mix—faculty and staff/admin making up 72.7% of the sample—likely influenced the types of questions posed and the expectations expressed. Faculty in particular appeared to test LINK against research-intensive queries, where database integration and specialized search capabilities are critical. By contrast, staff and students were more inclined to explore service-oriented questions, where LINK's scope is a better match. These specific queries also contributed to low ratings on the chatbot"s accuracy (18.2%).Usage patterns also suggest that this cohort is technologically engaged: over half regularly use virtual assistants and smart home devices, and 63.6% report routine use of other automated or AI assistants. At the same time, library resource use varied, with 36.4% accessing online resources less than once a week and only 27.3% doing so daily indicating the need for LINK to serve both high-frequency and occasional library users effectively.

Alignment with existing literature

These findings parallel a well-established body of research on library chatbots. Prior studies consistently point to the strong usability and around-the-clock availability that such systems provide, while also underscoring persistent gaps in user trust—particularly when it comes to accuracy, depth, and the ability to handle complex queries. Much of the literature highlights a recurring tension between user expectations and actual system capabilities. Once patrons encounter a conversational interface, their expectations tend to escalate rapidly, with many assuming the chatbot can function as a "full librarian", complete with access to subscription databases, catalog records, and personalized recommendations. Feedback from LINK reflects this same dynamic, reinforcing the conclusion that expectation management is just as critical as technical refinement. Clear interface cues, proactive disclosures, and guided query routing may ultimately prove as important as backend enhancements in shaping user satisfaction and sustaining trust.

Implications for system development

Based on the quantitative and qualitative findings, several priorities emerged for LINK's future development and integration. Reducing the database access gap remains essential, particularly by introducing anticipatory messaging when a query cannot be fulfilled due to content restrictions and providing seamless handoff pathways to catalog tools or human assistance. Expanding and optimizing the knowledge base, especially around high-demand topics such as research guides, course reserves, and core library services, will further strengthen perceived usefulness and maintain user trust. Continued accuracy improvements are also necessary, including the incorporation of more structured, source-linked answers that blend internal content with vetted external sources. Finally, maintaining a strong human connection is critical, as many users either prefer librarian expertise or do not fully trust AI to manage complex and nuanced questions. LINK should therefore function as a facilitator that guides users to library professionals when queries exceed its scope rather than attempting to replace human reference entirely.

LINK AI chatbot structure and organization

Iterative development and feedback sessions steered the development team toward a hybrid model chatbot that incorporated conversation pathing, similar to the chatbot models or weak AI documented in the literature, in combination with natural language response enabled by generative AI capabilities in Microsoft Copilot, trained on curated collections of library information. This approach was novel to the USF Copilot ecosystem and enabled a custom-built, dynamic model for the library to work with directly.

Fig. 8 illustrates a highly simplified conversational pathway within LINK. Maintaining connection to librarian assistance was paramount. Options to connect with the AskALibrarian service are provided at the beginning of each chat interaction. At the end of each user interaction, LINK requests user feedback. Depending on the user response, LINK offers to try again or suggests connection with a librarian through the

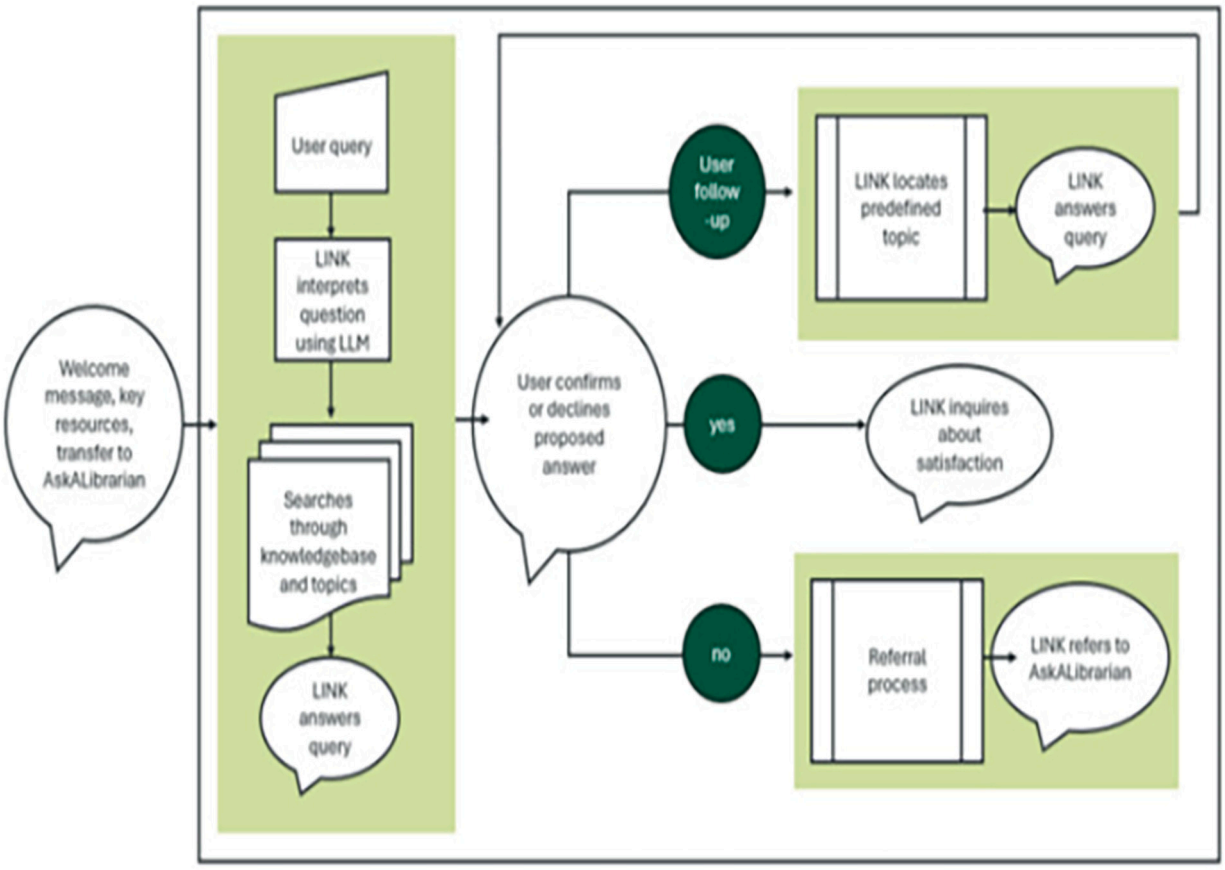


Fig. 8. LINK conversational pathway flowchart.

AskALibrarian service. The user can ignore the feedback question from LINK and input additional questions if desired.

Beta deployment

Beta deployment on the Libraries' website integrated LINK with the pre-existing, librarian staffed AskALibrarian chat. During open hours, library users are given the choice to initiate an interaction with LINK or to go directly to AskALibrarian. During closed hours, LINK provides immediate assistance when library personnel are not available. LINK introduces itself in each new chat, linking to a public information page on the Libraries' AI Tools and Resources guide: guides.lib.usf.edu/AI, and qualifying that it cannot access the internet.

Since LINK's beta launch in June 2025, the development team has issued weekly updates to address issues like hallucinations, broken links, and confusing logic patterns. A "report a chatbot problem" form has proven effective for routing and resolving issues quickly. Transcript reviews continue to identify subtle failures in logic or knowledge depth, often missed by users. In these cases, LINK's internal feedback loop helps flag questionable responses and improve them over time. By allowing the model more time in front of real patrons, the model has been exposed to prompts which expose it's knowledge gaps for the team. As a result, the Beta deployment proved essential to keep hallucinations at the minimum by full launch.

Full launch

The Fall 2025 semester (August 2025) marked the full launch of LINK as an integrated component of the USF Libraries' research support ecosystem. While the core functionality of the chatbot remained consistent—continuing to provide after-hours assistance and serving as an opt-in support tool for patrons—the transition from pilot to production shifted the team's focus from development to long-term operational stewardship.

With the conclusion of the initial grant-funded implementation period, maintenance became less intensive but more strategic. Post-launch work emphasized monitoring user transcripts, identifying recurring failure points, and proactively mitigating hallucination risks before they were encountered by patrons. Copilot Studio's recently introduced *Evaluation* tool supported this effort by enabling the team to run batches of test prompts outside of the live environment, allowing for structured quality review and iterative refinement.

As part of this monitoring process, the team also paid close attention to the risk of hallucinated or inaccurate responses. Notably, LINK exhibited very few instances of fully fabricated or overtly incorrect content in it's post-launch phase. Instead, the more common challenge involved the model offering generalized descriptions of library services or research guidance in cases where specific details were not available within its knowledge base. To address this, ongoing refinement efforts focused less on correcting misinformation and more on improving the specificity and structure of the information provided to the model. This included reorganizing pre-existing library content into more machine-readable formats, as well as regularly updating time-sensitive materials such as service schedules, semester-based policies, and evolving research support guidance. These practices reinforced the importance of transparency and referral to human expertise when uncertainty is detected. At the same time, they highlight an important operational reality: sustaining an AI-enabled reference service requires continual awareness of the broader information ecosystem of the library, and maintaining alignment between the chatbot's knowledge base and the library's frequently changing service landscape can be resource-

intensive without clearly mapped documentation workflows.

Although the original project goals centered on full deployment and reporting outcomes to library leadership, several ongoing priorities emerged as essential next steps for sustaining and expanding LINK's usefulness. Both the IRB study and the internal team discussions identified these trends concurrently. First, users consistently expressed interest in locating and accessing library materials directly through the chatbot interface, suggesting that tighter integration with collections systems could significantly enhance patron value. Second, key institutional repositories of information—particularly Springshare-hosted platforms such as LibGuides and LibCal—remained inaccessible due to vendor-imposed AI restrictions. Finally, continued updates to LINK's custom-built knowledge base were identified as necessary to ensure content relevance over time.

In response to these needs extending beyond the original grant period, library leadership allocated limited funding to support ongoing operations through the hiring of a student worker (15 hours per week). This role contributed not only to maintaining and expanding the knowledge base, but also to improving documentation workflows and establishing standards for future knowledge additions. Beginning in Spring 2026, the student worker also initiated development of sub-agents within LINK to further compartmentalize system functions and explore the potential for API-enabled connections to databases available to USF patrons. As of April 2026, the model now has limited Ex Libris Search REST API access and Elsevier Digital Commons RESTful API access. Full deployment is expected in Summer 2026, where patrons will be able to conduct searches through the LINK reference model.

Library leadership continues to engage with Springshare regarding AI access restrictions, though such limitations remain difficult to circumvent at present. The gap in AI-accessible content within current library information eco-systems may need to be considered more aggressively. While locally developed AI models are one issue, many patrons may be trying to learn about library services and best practices through AI models that crawl webpages. If vendor-hosted pages are blocking users' AI models, those models may end up relying on either incorrect information or even consider other hosted guidance from other institutions. (ex. Citation guidance from PurdueOWL instead of your own institution).

Post-launch usage and service impact

During the Fall 2025 semester, LINK handled a total of 247 chat sessions, 221 of which represented active dialogues between patrons and the model. Notably, the majority of these interactions occurred outside of staffed service hours, underscoring LINKs role as an after-hours extension of reference support. During the same period, the library received 306 chat interactions through the traditional Ask-A-Librarian service.

Taken together, LINK accounted for approximately 44% of all chat-based reference inquiries during the semester. However, overall chat volume across both services declined by 28% compared to the previous year. At this stage, the team cannot conclusively determine the cause of this net decrease. To ensure that LINK's always-available widget is not unintentionally suppressing use of Ask-A-Librarian during staffed hours, the team has begun developing a unified "dynamic single-container" widget model. This approach would dynamically inject Ask-A-Librarian into the widget between 10 am–5 pm and inject LINK during after-hours periods, rather than relying on external referrals to AskALibrarian through LINK during the day. Future assessment will examine whether this configuration better supports balanced engagement across services.

Sustainability and future directions

Overall, LINK has been viewed as a successful and sustainable service innovation. Library leadership has continued to authorize its use each semester and has consistently supported ongoing maintenance costs. Long-term operational expenses are expected to remain modest—approximately $232.50 per month under the library's contract with USF Research IT through a Citizen Developer support model.

Given LINK's contribution to nearly half of all chat reference interactions during the semester, these costs are outweighed by the benefit of maintaining a 24/7 research support platform that integrates with existing library services. With anticipated future enhancements—including API-based database searching, improved search strategy support, and task-specific sub-agents—LINK represents a scalable foundation for continued development of AI-enabled reference services in academic libraries.

Conclusion

Multiple rounds of user feedback and testing were essential in building an accurate, flexible library chatbot capable of extending services and improving efficiencies across the USF Libraries. From the outset, the development team prioritized transparency as a core design value to maintain patron trust. This approach was implemented through clear self-disclosure within the chatbot and reinforced by supplementary information pages. Although the IRB study was limited by a small respondent pool, the iterative feedback gathered throughout testing significantly shaped the model's development.

The resulting product, LINK, now functions as an additional pathway for navigating library resources while also generating usage data that will guide future research and assessment alongside traditional service points. Early deployment also highlighted the importance of ongoing maintenance, including keeping time-sensitive service information current and ensuring that responses remain aligned with library guidance and policies. Subsequent deployments have highlighted an adaptive development process post-launch, positioning LINK to grow beyond its initial launch state. Usage patterns will be reviewed each semester to ensure they are meeting user needs, and surveying or focus groups may be warranted as the model evolves to build upon the findings from the pre-launch IRB participant pool.

Despite encountering various friction points and unforeseen challenges during planning and development, the LINK team adapted effectively and delivered a tool that continues to enhance the overall success of the USF Bulls community. As LINK moves forward, it offers a sustainable foundation for continued exploration of how AI-enabled services can responsibly complement and extend academic library support.

CRediT authorship contribution statement

Evan Fruehauf: Writing – review & editing, Writing – original draft, Project administration, Investigation, Funding acquisition. **LeEtta Schmidt:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Amanda Boczar:** Writing – review & editing, Writing – original draft, Investigation, Data curation, Conceptualization.

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Declaration of competing interest

The authors declare no conflicts of interest.

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Appendix A. Survey instrument

Section 1. Participant background

1. What is your role at the university?
   - Undergraduate student
   - Graduate student
   - Faculty
   - Staff/Administration
   - Not affiliated with USF
   - Other (please specify).
2. What is your college or department?
   - Drop-down list of USF colleges/departments.
3. Which of the following technologies do you use on a monthly basis? (Select all that apply)
   - Virtual assistants (e.g., Siri, Alexa)
   - Recommendation systems (e.g., Amazon top picks)
   - Autonomous vehicles
   - Smart home devices (e.g., thermostats, appliances, lights)
   - Social media recommendations/algorithms
   - Smartwatch or wearable
   - Security cameras with face recognition
   - None of the above
   - Other (please specify).
4. How often do you use the library's online resources?
   - Daily
   - Several times a week
   - About once a week
   - Less than once a week
   - Rarely
   - Never.

Section 2. Library Use and Chatbot Interaction.

5. Which of the following library services have you used? (Select all that apply)
   - Online databases (e.g., JSTOR, PubMed, ProQuest)
   - *E*-books
   - Digital collections
   - Research guides
   - Reference/consultation services
   - Study room reservations
   - Workshops/instructional sessions
   - Printing/copying services
   - Course reserves
   - Interlibrary loan
   - Institutional repository, publishing, or data management services
   - Borrowing physical books or equipment
   - Special collections
   - None of the above; I was not aware of these services
   - Other (please specify).
6. What was your primary goal when using the library AI chat assistant?
   - Finding quick answers to basic questions
   - Getting help with a specific research query
   - Receiving personalized recommendations for resources
   - Learning more about library services
   - Other (please specify).
7. What was your initial impression of the library AI chat assistant? (Open-ended).

Section 3. Attitudes Toward AI and Chatbot Experience.

8. Please indicate your level of agreement with the following statements:

   Response options: Strongly disagree, Disagree, Neither agree nor disagree, Agree, Strongly agree

   a. I routinely use other automated or AI assistants.

   b. My past interactions with automated customer service chatbots have been helpful.

   c. I sometimes prefer automated assistants or chatbots for certain types of interactions.

   d. The library AI chat assistant was easy to use.

   e. The library AI chat assistant's responses were accurate.

   f. The library AI chat assistant understood my questions.

   g. I am generally satisfied with the library AI chat assistant.

   h. I will continue to use the library AI chat assistant in the future.

   i. I will recommend the library AI chat assistant to others.

   j. It is important to me that the library chat assistant is powered by AI.

   k. The library AI chat assistant is valuable for my academic or professional work.
9. What features of the library AI chat assistant did you find most useful? (Select all that apply)
   - Quick answers to basic questions
   - Ability to ask follow-up questions
   - Personalized recommendations
   - 24/7 availability
   - Other (please specify).
10. Can you describe a situation where the library AI chat assistant provided an incorrect or unhelpful response? (Open-ended).
11. What would you like to see improved in the library AI chat assistant? (Select all that apply)
    - More accurate responses
    - More personalized recommendations
    - Easier navigation
    - More comprehensive knowledge base
    - Other (please specify).

Section 4. Open-Ended Feedback and Follow-Up.

12. What do you think is the most significant area for improvement in the library AI chat assistant? (Open-ended).
13. How did the library AI chat assistant affect your overall experience with the library's online resources? (Open-ended).
14. Do you have suggestions for new features, functionalities, or topic areas to add to the library AI chat assistant? (Open-ended).
15. What do you think is the most significant benefit of using the library AI chat assistant? (Open-ended).
16. Do you have any additional comments or feedback about the library AI chat assistant? (Open-ended).
17. Do you want to be removed from all future communication about the library AI chat assistant?
    - Yes
    - No.

Data availability

Data will be made available on request.

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